

Invasive Plant Survey - Belt Transect Protocol

This protocol will be followed for surveying invasive plants near trails in Reno, Nevada. Each student group will survey the presence and absence of 23 species invasive to the Great Basin at a sampling site location using the *belt transect* method. At each sampling site location, 3 survey collection points will be selected along a trail by the student group, which will serve as the starting points for the 3 *belt transects*. Belt transects will be 30 meters in length and 6 meters in width, divided into 1 meter by 1 meter sampling squares (see *figure 1*). Students will use pacing to measure the 30 meter length of the transect and a tape measure to mark the width. Students will be provided a plant ID list to help identify plant species that are present in each sampling square, and a data sheet will be used to record the data.

1. At the sampling site, sample three x 30 meter *belt transects* with a width of 6 meters total (3 meters to the left of the trail edge and 3 meters to the right of the trail edge perpendicular to the center line).
2. Students will receive a sampling site location, and start and end points within the site, which will determine the boundaries for the 3 belt transects.
3. Students will choose 3 survey collection points for the 3 belt transects on the trail at the site. The 3 survey collection points have to be within the boundaries of the start and end points.
4. Measure 30 meters from the survey collection point parallel to the trail edge.
5. If the trail appears straight, use the 5 meter tape measure to measure your 30 meter transect line. If your trail curves, then you will use *pacing*.
6. Flag (or mark) the start and end points of your transect clearly.
7. For each 1 meter of the 30 meter center line, measure 3 meters to the left (from the left trail edge) and 3 meters to the right (from the right trail edge), perpendicular to the center line. For every 1 meter of the center line that is surveyed, 3 x 1² meter sampling squares to the left of the left trail edge will be surveyed, and 3 x 1² meter sampling squares to the right of the right trail edge will be surveyed. This is clearly represented on the data sheet.
8. Within each 1 x 1 meter sampling square, record on the data sheet which species (from the list of 23 target species) are present. A species is considered present if any portion of its aerial canopy is within the sampling square. Record all species that canopy overlaps the sampling square, including understory or groundcover species.
9. Within the correlating cell on the data sheet record plant species code for species present within sampling square (see species list), NS (for not surveyed), or a blank cell (for unrecorded).
10. For photo documentation: Take four photos for each transect.
 - a. Take a photo (landscape, 1x zoom) from the 0m mark looking up the transect line.
 - b. Take a photo (landscape, 1x zoom) from the 30m mark looking down the transect line.
 - c. Take a photo (landscape, 1x zoom) from the center line looking left
 - d. Take a photo (landscape, 1x zoom) from the center line looking right
11. Input plant data and photo documentation into the web app. If recording on a physical data sheet, transcribe the data from the data sheet to the web app after the survey is complete.

Definitions

- *Belt transect*: A monitoring technique used to measure frequency, and presence/absence of plant or animal species along a transect line using quadrats.

- [illegible]

Figure 1: This table demonstrates the layout of the belt transect. The 30 meter transect line will be measured on the trail. The width of the transect belt will be measured 3 meters to the right of the right trail edge and 3 meters to the left of the left trail edge perpendicular to the trail. Each sampling square will be 1 meter by 1 meter.